Week 7 Practice

```
library(haven)
library(labelled)
library(tidyverse)
library(tinytable)
```

```
upworthy <- read_csv("https://osf.io/vy8mj/download")
```

```
upworthy |>
  names()
```

```
 [1] "...1"                "created_at"          "updated_at"
 [4] "clickability_test_id" "excerpt"             "headline"
 [7] "lede"                "slug"                "eyecatcher_id"
[10] "impressions"         "clicks"              "significance"
[13] "first_place"         "winner"              "share_text"
[16] "square"              "test_week"
```

```
upworthy |>
  head()
```

```
# A tibble: 6 x 17
   ...1 created_at          updated_at          clickability_test_id     excerpt
  <dbl> <dttm>              <dttm>              <chr>                    <chr>
1    11 2014-11-20 11:33:26 2016-04-02 16:25:54 546dd17e26714c82cc00001c Things~
2    12 2014-11-20 15:00:01 2016-04-02 16:25:54 546e01d626714c6c4400004e Things~
3    13 2014-11-20 11:33:51 2016-04-02 16:25:54 546dd17e26714c82cc00001c Things~
4    14 2014-11-20 11:34:12 2016-04-02 16:25:54 546dd17e26714c82cc00001c Things~
5    15 2014-11-20 11:34:33 2016-04-02 16:25:54 546dd17e26714c82cc00001c Things~
6    16 2014-11-20 11:34:48 2016-04-02 16:25:54 546dd17e26714c82cc00001c Things~
# i 12 more variables: headline <chr>, lede <chr>, slug <chr>,
```

```
#   eyecatcher_id <chr>, impressions <dbl>, clicks <dbl>, significance <dbl>,
#   first_place <lgl>, winner <lgl>, share_text <chr>, square <chr>,
#   test_week <dbl>
```

```
upworthy_restricted <-
  upworthy |>
  select(
    created_at, clickability_test_id, headline, impressions, clicks
    )
```

```
head(upworthy_restricted)
```

```
# A tibble: 6 x 5
  created_at          clickability_test_id     headline       impressions clicks
  <dttm>              <chr>                    <chr>                <dbl>  <dbl>
1 2014-11-20 11:33:26 546dd17e26714c82cc00001c Let's See … H~        3118      8
2 2014-11-20 15:00:01 546e01d626714c6c4400004e People Sent T~        4587    130
3 2014-11-20 11:33:51 546dd17e26714c82cc00001c $3 Million Is~        3017     19
4 2014-11-20 11:34:12 546dd17e26714c82cc00001c The Fact That~        2974     26
5 2014-11-20 11:34:33 546dd17e26714c82cc00001c Reason #351 T~        3050     10
6 2014-11-20 11:34:48 546dd17e26714c82cc00001c I Was Already~        3061     20
```

```
upworthy_restricted <-
  upworthy_restricted |>
  mutate(
    exclamation =
      str_detect(string = headline, pattern = "\\?")
    )

upworthy_restricted |>
  count(exclamation)
```

```
# A tibble: 2 x 2
  exclamation     n
  <lgl>       <int>
1 FALSE       89559
2 TRUE        15992
```

```
exclamation_or_not <-
  upworthy_restricted |>
  summarise(
```

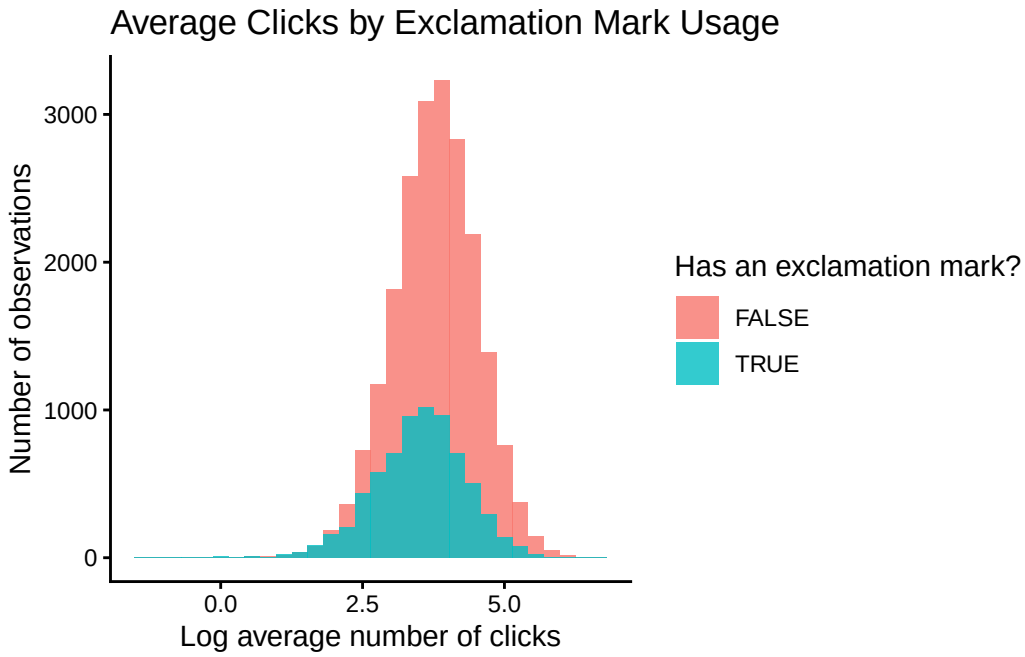
Exclaims!	Mean clicks
TRUE	45
FALSE	57

```
    ave_clicks = mean(clicks),
    .by = c(clickability_test_id, exclamation)
  )

exclamation_or_not |>
  pivot_wider(names_from = exclamation,
              values_from = ave_clicks,
              names_prefix = "ave_clicks_") |>
  drop_na(ave_clicks_FALSE, ave_clicks_TRUE) |>
  mutate(difference_in_clicks = ave_clicks_TRUE - ave_clicks_FALSE) |>
  summarise(average_differce = mean(difference_in_clicks))
```

```
# A tibble: 1 x 1
  average_differce
             <dbl>
1            -4.16
```

```
exclamation_or_not |>
  summarise(mean = mean(ave_clicks),
            .by = exclamation) |>
  tt() |>
  style_tt(j = 1:2, align = "lr") |>
  format_tt(digits = 0, num_fmt = "decimal") |>
  setNames(c("Exclaims!", "Mean clicks"))
```

```
exclamation_or_not |>
  ggplot(aes(
    x = log(ave_clicks),
    fill = exclamation
  )) +
  geom_histogram(
    bins = 30,
    position = "identity",
    alpha = 0.8
  ) +
```

```
  labs(
    title = "Average Clicks by Exclamation Mark Usage",
    x = "Log average number of clicks",
    y = "Number of observations",
    fill = "Has an exclamation mark?"
  ) +
  theme_classic()
```